

Quiescent or dusty? *Unveiling the nature of extremely red galaxies at $z > 3$*

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ABSTRACT

The advent of the *JWST* has revolutionized our understanding of high-redshift galaxies. In particular, the NIRCам instrument on-board *JWST* has revealed a population of red galaxies that had largely evaded detection with *Hubble Space Telescope* (*HST*), potentially due to significant dust obscuration, quiescence, or extreme redshift. Here, we present the first NIRSspec spectra of 23 red, *HST* faint or dark galaxies ($H - F444W > 1.75$), unveiling their nature and physical properties. This sample includes both dusty and quiescent galaxies with spectroscopic data from NIRSspec/PRISM, providing accurate spectroscopic redshifts with $\bar{z}_{\text{spec}} = 4.1 \pm 0.7$. The spectral features demonstrate that, while the majority of red galaxies are dusty, a substantial fraction, $13^{+9}_{-6}\%$, are quiescent. For the dusty galaxies, we have quantified the dust attenuation using the Balmer decrement ($H\alpha/H\beta$), finding attenuations $A_V > 2$ mag. We find that red dusty galaxies are $H\alpha$ emitters with equivalent widths spanning the range $68\text{\AA} < EW_{H\alpha} < 550\text{\AA}$, indicative of a wide range of recent star-formation activity. Whether dusty or quiescent, we find that red galaxies are predominantly massive, with 85 per cent of the galaxies in the sample having masses $\log(M_*/M_\odot) > 9.8$. This pilot NIRSspec programme reveals the diverse nature of *HST*-dark galaxies and highlights the effectiveness of NIRSspec/PRISM spectroscopic follow-up in distinguishing between dusty and quiescent galaxies and properly quantifying their physical properties. Upcoming research utilizing higher-resolution NIRSspec data and combining *JWST* with ALMA observations will enhance our understanding of these enigmatic and challenging sources.

Key words: techniques: spectroscopic – galaxies: high-redshift – infrared: galaxies.

1 INTRODUCTION

Over the past decade, infrared observations using the *Spitzer Space Telescope* (e.g. Caputi et al. 2012; Wang et al. 2016; Alcalde Pampiega et al. 2019) and mm-wavelength detections in Atacama Large Millimetre/submillimetre Array (ALMA) continuum data (e.g. Franco et al. 2018; Wang et al. 2019; Williams et al. 2019; Yamaguchi et al. 2019; Gruppioni et al. 2020; Xiao et al. 2023) have hinted at a significant population of dust-obscured galaxies that were missing from *Hubble Space Telescope* (*HST*) data sets. These objects have become known as ‘*HST*-dark’ galaxies. However, due to the constraints imposed by low spatial resolution observations (in the case of *Spitzer*) and the absence of spectroscopic redshifts, the reliable determination of their physical properties, including stellar

masses and star formation rates (SFRs), have remained extremely difficult. More recent studies with the *JWST* have now confirmed that massive, dusty, high-redshift galaxies were indeed missed from earlier (primarily optically selected) samples (e.g. Nelson et al. 2023; Barrufet et al. 2023a; Gómez-Guijarro et al. 2023; Labbé et al. 2023; Pérez-González et al. 2023; Rodighiero et al. 2023).

The high-resolution multiband $\lambda = 1\text{--}5\ \mu\text{m}$ photometry provided by *JWST*/NIRCам is transforming our understanding of these red, *HST* faint, or dark galaxies. Expressly, it has now been confirmed that these sources are generally located at redshifts $z > 3$ and typically lie on the main sequence of star-forming galaxies (e.g. Barrufet et al. 2023a; Rodighiero et al. 2023). Substantial further progress is now also possible by using the spectroscopic capabilities of *JWST*/NIRSspec to probe the physical properties of red galaxies in more detail (Williams et al. 2024; Pérez-González et al. 2024).

A further discussion triggered by early *JWST* data concerns the potential for the existence of massive galaxies at high redshift to

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present a challenge to the standard Λ CDM cosmological model (e.g. Boylan-Kolchin 2023; Lovell et al. 2023), although ongoing spectroscopic follow-up of several of the sources in question is already suggesting that such claims may have been premature. A notable case is that of CEERS-3210, a candidate early massive galaxy, initially reported as being at $z_{\text{phot}} \simeq 8$ by Labbé et al. (2023), which has now been revealed by NIRSpec spectroscopy to in fact be an active galactic nucleus (AGN) at $z_{\text{spec}} = 5.624$ (Kocevski et al. 2023). This example underscores the importance of securing spectroscopic redshifts for the accurate characterization of red galaxies in particular, given that the optical-infrared spectral energy distribution (SED) displayed by these sources often results in poorly constrained photometric redshifts, even with the best available NIRCам data.

The precise characterization of the number and nature of distant red galaxies is also of critical importance for the accurate determination of the high-mass end of the high-redshift galaxy stellar mass function (Gottmukkala et al. 2024, see also Weibel et al. 2024). While the current revolution in our understanding of early galaxy evolution is undoubtedly being driven primarily by *JWST*/NIRCам, it is already clear that imaging with the *JWST* Mid-Infrared Instrument (MIRI) camera at $\lambda > 5 \mu\text{m}$ is also invaluable. In particular, MIRI has been pivotal for confirming the presence of massive galaxies with significant dust attenuation at $z \simeq 7\text{--}8$ (Akins et al. 2023; Barro et al. 2024). Moreover, it has also been shown that the inclusion of MIRI data can significantly improve the constraints on inferred stellar masses, in some cases yielding more moderate values (Williams et al. 2024), which suggests that early galaxies do not present a challenge to our current understanding of cosmology (Wang et al. 2024). In parallel, Pérez-González et al. (2024) have underscored the importance of MIRI data for determining whether the continuum emission from red sources, including the compact objects now frequently referred to as ‘little red dots’ (LRDs; Matthee et al. 2024), is dominated by emission from an obscured AGN or starlight.

In addition to confirming the existence of dusty galaxies at high redshift, *JWST* is also revealing significant numbers of massive galaxies in the early Universe that are red because their stellar populations are already significantly evolved at $z \simeq 3\text{--}5$, implying that star formation activity in these galaxies was somehow terminated at very early times (e.g. Carnall et al. 2023, a, b; Valentino et al. 2023). These galaxies are red primarily because they exhibit prominent breaks in their spectra at rest-frame $\lambda_{\text{rest}} \simeq 4000 \text{ \AA}$. However, it can still be very challenging to distinguish them from dusty galaxies when deep imaging is only available at $\lesssim 2 \mu\text{m}$. While these objects often appear bluer at longer wavelengths, meaning that in principle, they can be distinguished from their dusty counterparts using NIRCам + MIRI photometry, recent findings also highlight the existence of galaxies at $z \simeq 4$ that are both quiescent and dusty (Setton et al. 2024). The accurate identification of quiescent galaxies at $z > 3$ is crucial for precisely determining their number density, and hence advancing our understanding of the onset and subsequent quenching of star formation activity at very early times (i.e. Merlin et al. 2019; Santini et al. 2021).

The advent of extensive *JWST* imaging surveys, as exemplified by the Public Release Imaging for Extragalactic Research (PRIMER) survey (PI: J. Dunlop), heralds a new era of statistically meaningful studies of red galaxies (both quiescent and dusty) due to the power of extensive NIRCам + MIRI imaging over the best-studied *HST* legacy fields (Barrufet et al. in preparation; Dunlop et al. in

preparation). However, even with expanded NIRCам + MIRI imaging data, it is clear that spectroscopic follow-up with *JWST*/NIRSpec will be crucial for the study of red galaxies in particular. Such observations are essential for delivering reliable and precise spectroscopic redshifts for such objects, breaking the photometric redshift degeneracies often arising from their red SEDs. Moreover, the broad wavelength coverage and moderate spectral resolution provided by the NIRSpec PRISM mode, in particular, provide both a full suite of emission lines (where present), as well as high signal-to-noise ratio (SNR) detections of continuum emission (and hence, potentially, absorption lines), enabling dusty and quiescent red galaxies to be unambiguously distinguished. NIRSpec follow-up spectroscopy is thus urgently required to resolve the outstanding questions posed by the latest advances in the study of red galaxies, which have arisen from the deep *JWST* imaging observations.

In this rapidly evolving context, we here report the first results from the Cycle-1 NIRSpec pilot programme, ‘Quiescent or dusty? Unveiling the nature of red galaxies at $z > 3$ ’ (Programme ID: 2198; PIs: L. Barrufet & P. Oesch), providing the first *JWST* spectroscopic follow-up data focused explicitly on red galaxies. For the reasons explained above, the key aims of this programme are to (i) deliver spectroscopic redshifts, (ii) enable the classification of individual red galaxies as either quiescent or dusty, and (iii) provide robust estimates of their fundamental physical properties, including dust attenuation and stellar masses.

The paper is structured as follows. In Section 2, we describe the design of our NIRSpec programme and provide details of the source catalogue produced from our NIRCам preimaging (through the F444W and F200W bands). Our multiband photometric catalogue is completed with photometry from First Reionization Epoch Spectroscopically Complete Observations (FRESCO; Oesch et al. 2023) and from the *JWST* Advanced Deep Extragalactic Survey (JADES; Eisenstein et al. 2023a). The methodology we adopted to analyse our spectroscopic data is presented in Section 3, and then our spectroscopic results are presented in Section 4, where a distinction between quiescent and dusty categories is made, and the results of this classification are discussed in the context of observed galaxy morphology. The derived physical properties of the red galaxies are presented in Section 5, and we then discuss our results in Section 6. Finally, we summarize our conclusions in Section 7.

Throughout, all magnitudes are given in the AB system (e.g. Oke & Gunn 1983), and for all cosmological calculations, we adopt $\Omega_{\text{M}} = 0.3$, $\Omega_{\Lambda} = 0.7$, and $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

2 OBSERVATIONS

As outlined above, the primary aim of our programme ‘Quiescent or dusty? Unveiling the nature of red galaxies at $z > 3$ ’ (GO-2198; PIs: L. Barrufet & P. Oesch) was to obtain low-resolution NIRSpec PRISM spectra of a representative sample of red galaxies to determine the redshifts and nature of these enigmatic galaxies. In this section, we describe how our targets were selected and summarize the basic reduction and analysis of the NIRSpec data, which cover an observed wavelength range $1\text{--}5 \mu\text{m}$ at a typical spectral resolution of $R = 100$.

As we describe further below, we were able to target 23 red galaxies that are the focus of this paper. However, our two NIRSpec pointings meant that we were able to obtain spectra of ~ 140 objects with the NIRSpec Micro-Shutter Assembly (MSA). To fully exploit this observational opportunity, we therefore selected ~ 110 ‘fillers’,

which we selected as potential massive galaxies at $z > 3$ from the 3D-*HST* catalogue (Skelton et al. 2014).

2.1 (Pre-)imaging data and photometric catalogue

Our NIRSpec programme was designed to follow up on several *HST*-dark galaxies that were initially identified in *Spitzer*/IRAC imaging and remained undetected in *HST* imaging data in the GOODS-S field from the GREATS survey (Labbé et al. 2015; Stefanon et al. 2021). The programme’s main targets could be covered with two pointings of NIRSpec. This programme ultimately included both *HST*-dark and *HST*-faint galaxies, which we have collectively rephrased as red galaxies for nomenclature clarity.

Since the IRAC point-spread function is much too wide to pinpoint the accurate location of these galaxies, *JWST* NIRCам pre-imaging was required. This pre-imaging was taken between 2022 November 10 and 19 in the F444W and F200W filters. The short exposures were taken with the BRIGHT2 readout pattern with seven groups/integration and a six-point FULLBOX dither pattern, leading to a total exposure time of 901 s. For each of the two pointings, the images were mosaicked in two rows to span the entire field of view of NIRSpec.

The images were processed and aligned with the *grizli* tool before deriving multiband photometric catalogues. In addition to the *JWST*/NIRCам images, pre-existing *HST* imaging with ACS and WFC3/IR was included. Finally, a small portion of each field overlapped with FRESCO, where public F182M, F210M, and deeper F444W imaging were available for the catalogues used for creating the masks back in 2022 November. Since then, a good part of the field has additionally been covered by deep JADES imaging from the original GTO programme and the cycle 2 JADES Origins Field (Eisenstein et al. 2023a, b). Hence, several of our targets are now covered by deep multiband photometry, including medium-band filters.

For the photometric catalogues adopted in the following, we use all the v7 ACS, WFC3/IR, and NIRCам images available in the GDS-SW field on the public DAWN *JWST* Archive (DJA).¹ Since these do not yet include the pre-imaging from this programme, this was added in a separate stack of the F200W and F444W data. This ensures that all sources are covered by at least two NIRCам bands. All images were aligned to the same 40 mas pixel grid.

To construct photometric catalogues, we follow the same procedure as outlined in Weibel et al. (2024, see also e.g. Barrufet et al. 2023a; Gottumukkala et al. 2024). Briefly, SExtractor (Bertin & Arnouts 1996) is run in dual image mode using a point spread function (PSF)-matched, inverse-variance weighted stack of the F277W, F356W, and F444W images as the detection image. The remaining filters were PSF-matched to F444W using PSFs created directly from stars in the images. Photometry was measured in small circular apertures of 0.48 arcsec diameter and corrected to total fluxes using Kron AUTO apertures from the F444W images plus a small residual correction for flux lost from the wings of the PSF.

As a general quality cut, the catalogue only contains objects with $S/N > 5$ in at least one of the available NIRCам-wide filters. We also removed objects near the edge of the image, stars, and spurious detections. Finally, to account for possible systematic uncertainties in the photometry, we apply an error floor of 5 per cent to all flux measurements before running any SED fitting.

2.2 Spectroscopic data

This programme contains two NIRSpec pointings that allowed us to target 140 galaxies with PRISM spectra at $R \sim 100$. In each pointing, two MSA slit configurations were used for observations. These were observed on 2023 January 19 and 2023 February 6, respectively. The second pointing suffered NIRSpec shutter glitches affecting the spectra of the main sample, which was removed in this study. For each MSA mask, the spectra were taken with the PRISM grating and an NRSIRS2RAPID readout pattern of 55 groups/integration. We used three slitlets per target and dithered along the three shutters. Each spectrum thus received a total exposure time of 2451 s. This allows us to reach the continuum for faint sources and probes emission lines down to $\sim 1 \times 10^{-18} \text{ erg s}^{-1} \text{ cm}^{-2}$ (5σ).

The 140 NIRSpec spectra were reduced and extracted with *msaexp*,² a tool for extracting *JWST* NIRSpec MSA spectra. Individual slits were identified, considering the three shutter positions. After subtracting the sky background, the data were collapsed into a single 2D spectrum. This was then processed to extract the 1D spectra. Our data are identical to that available online on the DJA,³ except for source ID = 7151, which was processed manually for improved slit extraction.

Spectroscopic redshifts were also determined using the MSAExp tool. The wide wavelength range covered by the NIRSpec/PRISM spectra ($\sim 1\text{--}5 \mu\text{m}$) allowed us to determine an accurate spectroscopic redshift for every source in our mask, as they encompassed several spectral lines or the continuum with a Balmer Break for all sources.

The 2D and 1D spectra of the primary sample of this study, including their IDs and the spectroscopic redshifts of the *HST* dark galaxies, are presented in Appendix A.

2.3 Selection of red galaxies

The primary targets of this programme are red galaxies. However, the masks were filled with other galaxies based on previous catalogues from 3D-*HST* (Skelton et al. 2014) and our pre-imaging data. The total NIRSpec sample of 140 sources was classified by red colour using the *HST* H-band and the reddest band of NIRCам, F444W, at 1.6 and $4.4 \mu\text{m}$, respectively. Using a similar colour cut as employed in Barrufet et al. (2023a) of $H\text{-F444W} > 1.75$, a total of 23 red galaxies in our MSA masks. These are highlighted in Fig. 1. Our primary targets span a broad range in F444W magnitude between 22 and 26 mag. While these sources would be bright enough to be identified with *Spitzer*/IRAC data in these fields (e.g. Stefanon et al. 2021), only three among these sources were present in the previous catalogues of H-dropout galaxies from Wang et al. (2019). In large part, this can be attributed not only to the $\sim 10\times$ lower spatial resolution of the IRAC instrument compared to *JWST*/NIRCам data but also to a redder colour cut of $H\text{-}[4.5]_{\text{F444W}} \gtrsim 2$ in *Spitzer*-based *HST*-dark samples (e.g. Wang et al. 2016).

With the NIRSpec spectra in hand, we can finally study the nature of these galaxies, which we do in the following sections. In Fig. 1, we already highlight three distinct types of galaxies that are identified among the red sample that will be introduced and discussed in more detail in Section 4.

¹<https://dawn-cph.github.io/dja/imaging/v7/>

²<https://github.com/gbrammer/msaexp/>

³<https://dawn-cph.github.io/dja/>

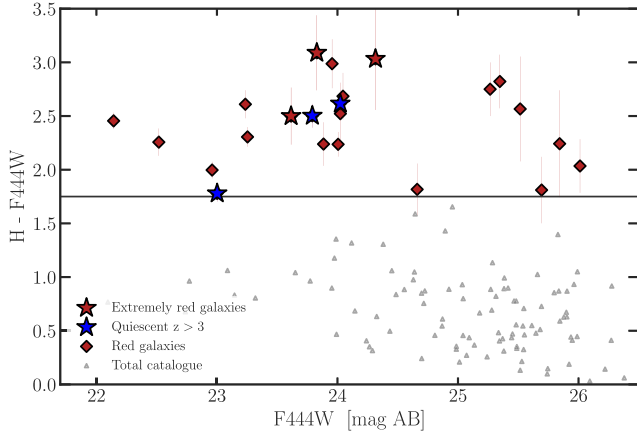


Figure 1. Colour–magnitude diagram (F444W versus H-F444W) of the total sample (grey triangles) of the ‘Quiescent or dusty?’ programme (GO 2198). We selected 23 red galaxies (red diamonds) with the colour H-F444W > 1.75 similar to Barrufet et al. (2023a) (black line). This subset includes several types of red galaxies: red stars represent those with extremely red spectra, while blue stars denote the three quiescent galaxies at $z > 3$, respectively indicating either significant dust presence or, conversely, a pronounced Balmer break.

3 METHODOLOGY

This section outlines the methodology employed to determine the physical properties of red galaxies. It uses spectroscopic data for classification and spectroscopic redshifts alongside photometric data for quantifying stellar masses, SFRs, and dust attenuation.

3.1 Emission-line extractions

We classified red galaxies by directly examining their spectra. Galaxies were identified as quiescent if they exhibited a clear Balmer break without significant emission lines, distinguishing them from those with steady dust attenuation. Conversely, galaxies were classified as ‘dusty’ if their spectra showed dust attenuation $A_V \gtrsim 1$ mag. The red spectral slope is another sign of dustiness, indicating significant attenuation.

The dusty red galaxies in our sample feature prominent $H\alpha + [N II]$ emission, with additional spectral lines like $[S II]$, $H\beta$, and $[O III]$ varying among galaxies. Hence, dust attenuation was assessed using the Balmer decrement between $H\alpha$ and $H\beta$, with spectroscopic redshifts obtained from PRISM spectral fitting (see Section 2.2). Utilizing these redshifts, we performed Gaussian fits to the $H\alpha$ and $H\beta$ lines to calculate their fluxes for dust attenuation assessment. More specifically, we simultaneously fit the spectral regions encompassing $H\beta + [O III]$ and $H\alpha + [N II] + [S II]$ using Gaussian profiles for each line. We use a 400 Å spectral region for the continuum on both sides of these lines. Fig. 2 shows an example of the best-fitting results for two of our targets, one dusty and a quiescent galaxy. However, when $H\beta$ lines remained undetected, we used a conservative 2σ upper limit. The continuum adjacent to these lines was fitted, and the line flux was subtracted. Following this subtraction, we applied the method described by Domínguez et al. (2013) to determine the colour excess, using $E(B - V) = 1.97 \log((H\alpha/H\beta)/2.86)$ based on a Calzetti et al. (2000) attenuation curve. We converted from colour excess to dust attenuation A_V with the conversion factor applicable for starburst $R_V = 4.05$.

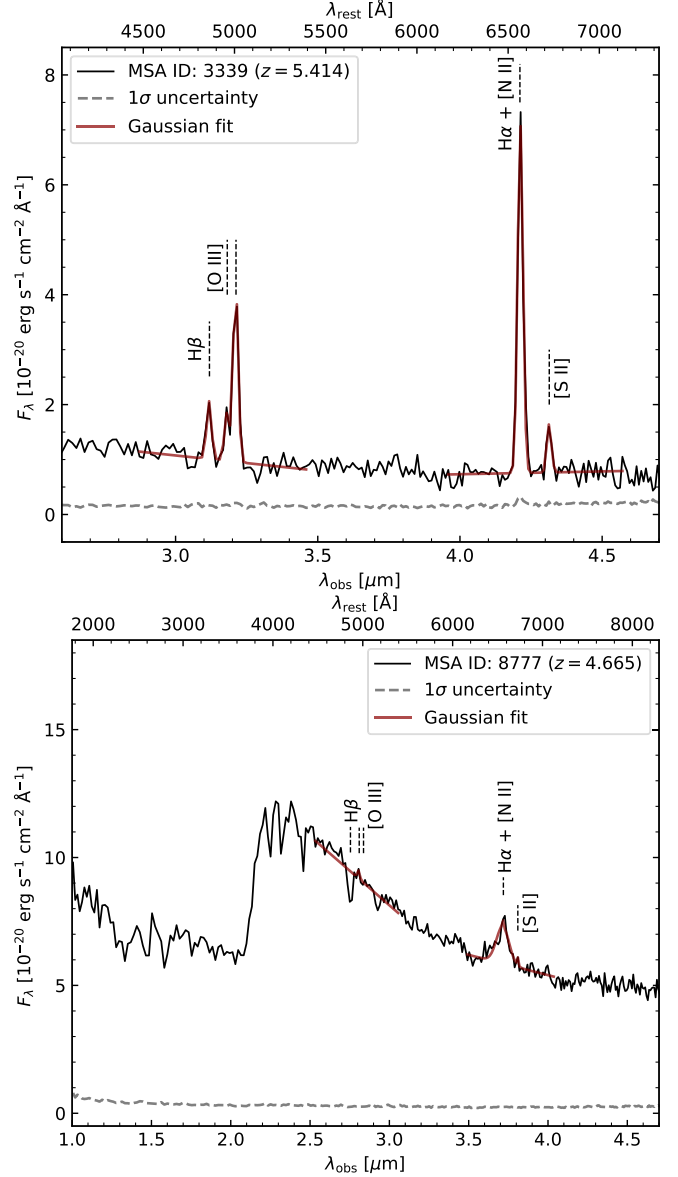


Figure 2. This figure presents spectral fitting examples for two distinct red galaxies: dusty and quiescent. Black lines denote the observed spectra, while the red lines depict Gaussian fits to spectral lines and the linear fitting to the continuum. Top: The dusty galaxy (ID = 3339) presents emission lines ($H\beta$, $[O III]$, $H\alpha + [N II]$, and $[S II]$). Bottom: The quiescent galaxy (ID = 8777) is characterized by its typical spectral profile and absence of emission lines, except for an unusual $H\alpha + [N II]$ emission because of the AGN presence.

Note as a caveat that, due to the PRISM’s resolution, $H\alpha$ and $[N II] \lambda 6584$ fluxes are blended, leading to an overestimation of $F_{H\alpha}$. To minimize this effect, we have inspected the contribution of $[N II]$ to $F_{H\alpha}$ by varying 10 per cent of its contribution (e.g. Sanders et al. 2015; Mármol-Queraaltó et al. 2016). We find that the values of A_V change up to 0.37 dex, with non-critical impact in the resulting observed fluxes of $H\alpha/H\beta$. However, follow-up high-resolution observations of red galaxies are essential to corroborate the contribution $[N II]$ to the flux measurement of $H\alpha$.

Additionally, for $H\alpha$ – the sole line present in all dusty red galaxies – we measured the line’s full width at half-maximum (FWHM) and the rest-frame equivalent width (EW). Given the low resolution of

the prism spectra, however, the FWHM are not conclusive of broad lines except for two sources (see the results in Section 4).

3.2 Spectral energy distribution fitting

We computed the physical properties of the 140 sources utilizing the multiband photometric catalogue described in Section 2 and incorporating the spectroscopic redshifts described in Section 2.2. Specifically, we determined their SFRs, levels of dust attenuation, and stellar masses, emphasizing the latter to evaluate if these sources are among the most massive observed in our sample. We apply the same methodology to red galaxies and the remaining sample, facilitating a robust comparison.

For this analysis, we employed the Bayesian Analysis of Galaxies for Physical Inference and Parameter ESTimation (BAGPIPES) tool (Carnall et al. 2018), which constructs complex model galaxy spectra and fits them to a combination of spectroscopic and photometric data using the MultiNest nested sampling algorithm (Feroz et al. 2019). Importantly, unlike prior NIRCcam studies, our approach benefits from incorporating spectroscopic redshifts as an input, thereby reducing the degeneracies often associated with dual peaks in photometric redshift distributions caused by dust attenuation effects on redshift determination.

We have adopted similar model assumptions to those in Barrufet et al. (2023a) and Gottumukkala et al. (2024), which have effectively characterized *HST*-dark and *HST*-faint galaxies. We employ a delayed star formation history (SFH) model with an e-folding time ranging from $\tau = 0.1$ –9 Gyr, using a uniform prior. This allows for the inclusion of both relatively short bursts and, effectively, a constant SFH. The stellar population models are based on the 2016 updated version of the Bruzual & Charlot (2003) library, using a Kroupa (2001) initial mass function (IMF). We consider a range of metallicities from 0.2 to 2.5 $Z_{\odot}$, taking the Solar value as $Z_{\odot} = 0.02$. Nebular continuum and emission lines are included self-consistently, utilizing the photoionization code CLOUDY (Ferland et al. 2017), with the ionization parameter set to $\log U = -2$. Lastly, we apply the Calzetti et al. (2000) dust attenuation law, accommodating highly attenuated SEDs by setting an A_V range of 0–6 mag, using a uniform prior.

4 RED GALAXIES SPECTROSCOPIC RESULTS: QUIESCENT, DUSTY, AND AGN?

This section presents a first spectroscopic analysis of red galaxies spanning the ~ 1 –5 μm wavelength range facilitated by *JWST*/NIRSpec observations. Additionally, we quantify the percentage of quiescent galaxies within our sample.

Four of 23 analysed red galaxies were excluded due to low-quality spectra, leaving 19 with high-quality spectra suitable for spectroscopic analysis. Despite these four spectra being compromised, noisy emission lines and photometric data still confirmed the galaxies' dusty nature.

Even with NIRCcam photometry, the photometric redshifts of red galaxies are difficult to determine due to the red SEDs of these galaxies. Therefore, the first question to address is simply the redshift distribution of these galaxies. This is shown in Fig. 3, juxtaposed against earlier *JWST*/NIRCcam photometric analyses. We find a median redshift of $\bar{z} = 4.1 \pm 0.7$, with all sources located at $z \gtrsim 3$. The results confirm the efficacy of red colour criteria for pinpointing red galaxies at $z > 3$, aligning with prior research that utilized NIRCcam photometry (e.g. Pérez-González et al. 2023;

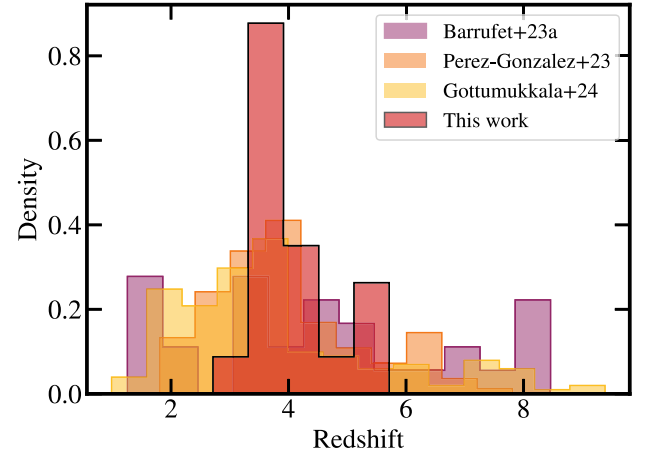


Figure 3. Redshift distribution of red galaxies: this histogram presents our spectroscopically derived redshifts (red distribution) alongside photometric redshifts from comparative studies from Pérez-González et al. (2023), Barrufet et al. (2023a), and Gottumukkala et al. (2024). Our data range from $z \sim 2.7$ –5.5 and peak at $z \sim 4$, underscoring our sample's mean redshift of red galaxies. Photometric data extend to higher redshifts beyond the scope of our pilot study, indicating the necessity for more in-depth observations.

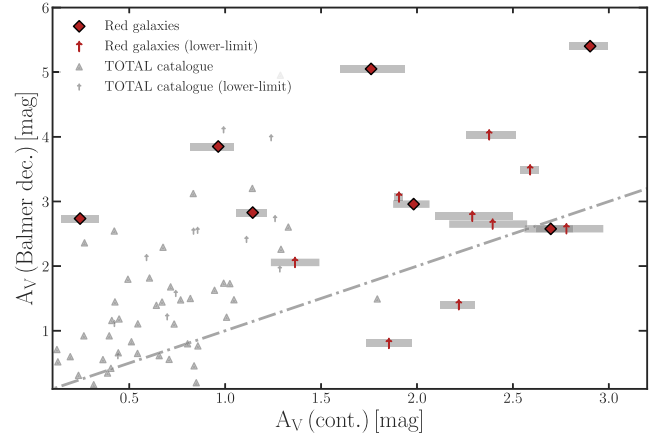


Figure 4. Dust attenuation comparison: Balmer decrement-derived A_V versus continuum-derived A_V . Red galaxies that are indicated by red diamonds (with $H\beta$ detection) and red arrows (2σ lower limit for non-detection; uncertainties shown by grey bars indicate the 16th and 84th percentiles) exhibit larger attenuation than the rest of the sample (non-red dark galaxies, grey triangles, and arrows). The dashed line presents the one-to-one relation. Red galaxies generally surpass $A_V > 3$ mag and even exceed $A_V > 5$ mag. While a correlation exists between spectral and continuum A_V magnitudes, spectral measurements are consistently higher than continuum. Not surprisingly, while spectral lines probe more dust-obscured regions than continuum, there could have been a potential underestimation of dust presence in red galaxies by using photometric observations.

Rodighiero et al. 2023; Williams et al. 2024; Barrufet et al. 2023a; Gottumukkala et al. 2024).

4.1 Are dusty red galaxies a homogenous population?

Despite indications of significant dust obscuration from NIRCcam studies, Balmer decrements are a much more direct way of estimating dust attenuation in galaxies. Fig. 4 compares red galaxies to the full sample of galaxies, for which we have obtained NIRSpec/PRISM

Table 1. Spectral properties of dusty red galaxies. The first column lists the galaxy IDs; the second and third columns display the coordinates, RA and Dec., respectively. The fourth and fifth columns present the fluxes of $H\alpha$ and $H\beta$, along with their uncertainties ($>H\beta$ flux indicates a 2σ upper limit). The sixth column details the attenuation derived from the Balmer decrement ($>$ indicates a lower limit). The seventh column shows the rest-frame EWs of the $H\alpha$ lines. The final column contains the spectroscopic redshifts (all the spectroscopic redshifts contain at least two lines except for ID = 1548 since the spectrum is cut). Notice that the $H\alpha$ values are blended with [N II].

ID	RA (deg)	Dec. (deg)	$F_{H\alpha}$ (10^{-20} erg s $^{-1}$ cm $^{-2}$)	$F_{H\beta}$ (10^{-20} erg s $^{-1}$ cm $^{-2}$)	A_V (mag)	EW $_{H\alpha}$ (Å)	z_{spec}
1260	53.0748575	−27.875898	290 ± 82	<68	>1.40	68 ± 26	4.44
1548	53.1347898	−27.907492	463 ± 44	<72	>2.77	70 ± 9	4.43
1616	53.0766784	−27.8734698	3553 ± 104	<452	>3.49	331 ± 17	3.46
3012	53.0767591	−27.8641199	414 ± 50	<115	>0.80	149 ± 44	3.47
3339	53.1147448	−27.8904202	1495 ± 50	230 ± 31	2.83	304 ± 59	5.41
3640	53.0821208	−27.8598697	720 ± 94	<78	>4.03	146 ± 31	3.65
4307	53.1406347	−27.8813756	1137 ± 55	83 ± 26	5.40	375 ± 231	5.52
4490	53.0410546	−27.8544706	1829 ± 72	290 ± 42	2.74	484 ± 81	3.70
4820	53.034904	−27.8521791	1966 ± 46	225 ± 36	3.85	545 ± 103	3.79
5446	53.0279874	−27.8470862	1261 ± 51	<209	>2.58	267 ± 24	2.71
5510	53.1783286	−27.8702734	1369 ± 49	<264	>2.06	134 ± 6	3.47
5607	53.1233683	−27.8705379	434 ± 28	72 ± 28	2.58	346 ± 38	3.59
6151	53.0420756	−27.8426437	316 ± 36	<51	>2.65	2025 ± 176	4.38
12469	53.0446935	−27.8136318	2177 ± 50	323 ± 63	2.96	280 ± 17	3.55
12577	53.0484576	−27.8151421	1799 ± 66	146 ± 19	5.05	977 ± 67	5.23
7151	53.05672	−27.836383	1427 ± 67	<119	>4.8	2647 ± 93	3.79

spectra. Red galaxies are among the most dust-enriched in the sample. Importantly, the attenuation measured from the Balmer decrement presents significantly larger values than that derived from SED fitting, with differences that vary considerably from galaxy to galaxy. This can be expected due to the inhomogeneous distribution of gas and dust (see e.g. Calzetti et al. 2000).

Table 1 presents the spectral parameters of red galaxies, such as fluxes and dust attenuation, along with $H\alpha$ EWs. The $E(B - V)$ ranges within 0.2–1.3 that translates to dust attenuations of $0.8 < A_V/\text{mag} < 5.4$ assuming $R_V = 4.05$ (Kashino et al. 2013). Among the galaxies with $H\beta$ detections, meaning that the attenuation is computed with the $H\beta$ flux and not a lower limit, the dust attenuation from Balmer lines is $A_V/\text{mag} \gtrsim 2.6$, indicating high attenuation in red galaxies. Red galaxies exhibit larger dust attenuation than the rest of the galaxy sample in our mask, as evidenced by both the continuum and the Balmer decrement.

Intriguingly, dusty red galaxies exhibit diverse spectral shapes, with a significant majority presenting red spectra. This observation prompted us to categorize them based on spectral profiles to understand their intrinsic properties better. Fig. 5 presents the reddest spectra, which align with the redder H-F444 colours. Notice that the shape of the spectra and the redshift ($z \sim 4.4$) are almost identical. IDs 1548 and 6151 exhibit substantial attenuation with $A_V > 2.2$ mag, whereas ID 1260 presents $A_V > 1.1$ mag. Due to the lack of $H\beta$ flux measurements, we cannot conclude that the galaxies have different dust attenuation. Moreover, these galaxies rank among the most massive in our data set, akin to quiescent galaxies at similar redshifts (see physical properties in Section 5). As expected, the (F160W–F444W) colour, while a proxy for the mass-to-light ratio (and thus indirectly correlated with galaxy mass, as more massive galaxies tend to be redder), does not effectively distinguish between different galaxy types, such as dusty and quiescent galaxies, as discussed in Section 6.

In contrast to galaxies with red-sloped spectra, Fig. 6 shows red galaxies with flatter spectra and $H\beta$ emission, enabling direct dust attenuation calculations. Among the highest redshift galaxies in our sample ($z > 5$), these three exhibit varying dust attenuation levels.

Specifically, IDs 4307 and 12577 show substantial attenuation at $A_V/\text{mag} = 5.4$ and 5.1, respectively, in contrast to the less severe $A_V/\text{mag} = 2.8$ observed in ID 3339. Despite being dusty, these galaxies have moderately lower stellar masses ($\log(M_*/M_\odot) \sim 10$), 0.5 dex below the redder galaxies (see Section 5). We note that only source ID = 12577 presents an $H\beta$ line broader than [O III], which suggests the presence of an AGN (see Fig. 7).

Since all dusty red galaxies are $H\alpha$ emitters, we also analysed the rest-frame EW for the $H\alpha$ lines. We find a broad range of EW with $68 < \text{EW}/\text{\AA} < 545$ except for the ID = 12577 with an EW ~ 1000 Å. Assuming the blended $H\alpha + [\text{N II}]$ emission predominantly originates from H II regions, with [N II] contributing minimally (but in AGN, such a contribution could be notable), the inferred EWs indicate substantial recent star formation activity (within the last ~ 10 Myr). However, it is critical to note that our observations reveal substantial variations in attenuation levels between the nebular and the stellar continuum emission (see Fig. 4). This disparity introduces significant uncertainties in deducing SFHs based solely on EW $H\alpha$ measurements.

Our findings conclusively establish the dusty nature of most red galaxies through spectroscopic evidence, aligning with observations from NIRCcam dust continuum data. While most red galaxies do not show clear AGN features, we only have evidence for one source to harbour an AGN.

4.2 Quantifying the quiescent fraction in red galaxies

Recent *JWST* NIRCcam results suggest that red galaxies might not only be dusty star-forming galaxies but could also exhibit quiescent characteristics (Gómez-Guijarro et al. 2023). Notably, Pérez-González et al. (2023) estimated that quiescent galaxies may represent about 18 per cent of the *HST*-dark/faint galaxy population. This section delves into identifying and analysing quiescent galaxies within our red galaxy sample.

Our pilot study reveals that 3 out of the 23 red galaxies, i.e. 13^{+9}_{-6} per cent are quiescent. Fig. 8 presents the spectra of these three sources. The first spectrum (ID = 8777) at $z_{\text{spec}} = 4.7$ shows

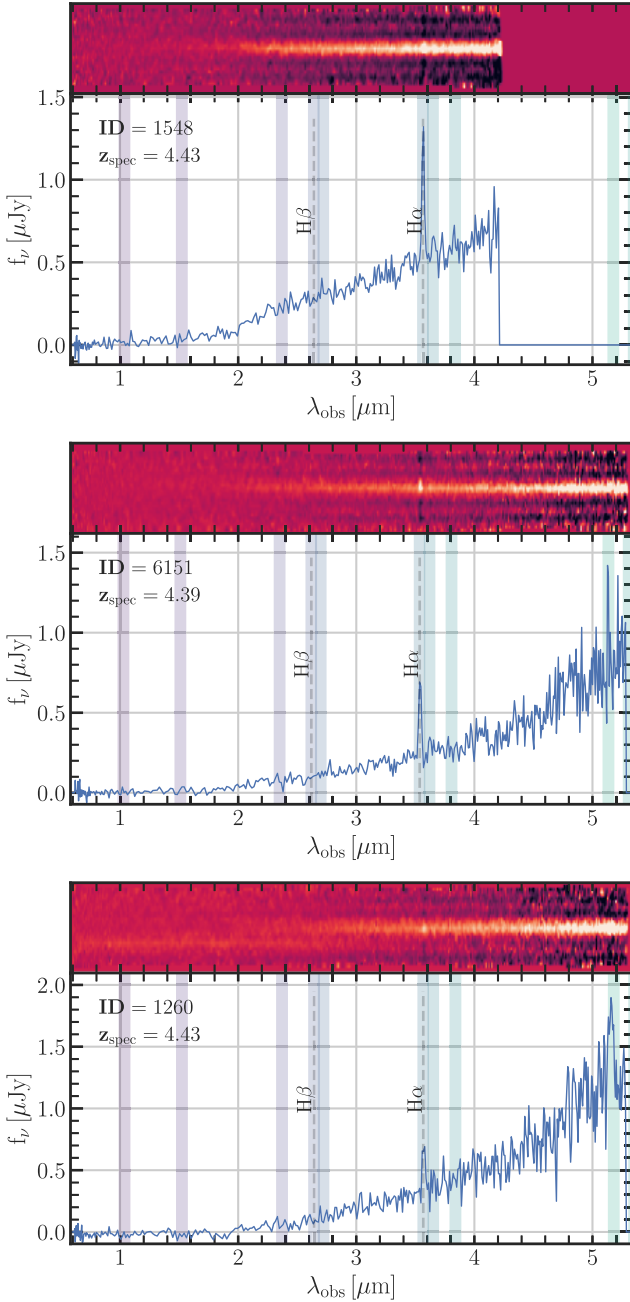


Figure 5. 2D and 1D spectra from three red galaxies with extreme red slopes. The dashed lines mark the locations of the $H\alpha$ and $H\beta$ spectral lines. Coloured lines represent [O III], He I, [C I], and [S II]. These galaxies display nearly identical spectra and lie at similar redshifts of $z \sim 4.4$. They show a distinct red slope with a prominent $H\alpha$ line but lack $H\beta$ (dashed lines) and [O III] lines. The adjacent [S II] line (in green) to the $H\alpha$ line and the detection of [S III] at $\lambda_{\text{obs}} \sim 5.2 \mu\text{m}$ (in both ID = 6151 and ID = 1260) support the accuracy of the redshift measurement.

the highest-redshift quiescent galaxy reported to date by Carnall et al. (2023a). The presence of a broad $H\alpha$ line with an observed $\text{FWHM} = 5720 \pm 654 \text{ km s}^{-1}$ indicates AGN activity, despite the low equivalent width ($\text{EW} = 32 \pm 5 \text{ \AA}$).

The second quiescent galaxy (ID = 8290) is at a marginally lower redshift of $z_{\text{spec}} = 4.4$, in agreement with the photometric redshift reported by Carnall et al. (2020). The third spectrum (ID = 6620)

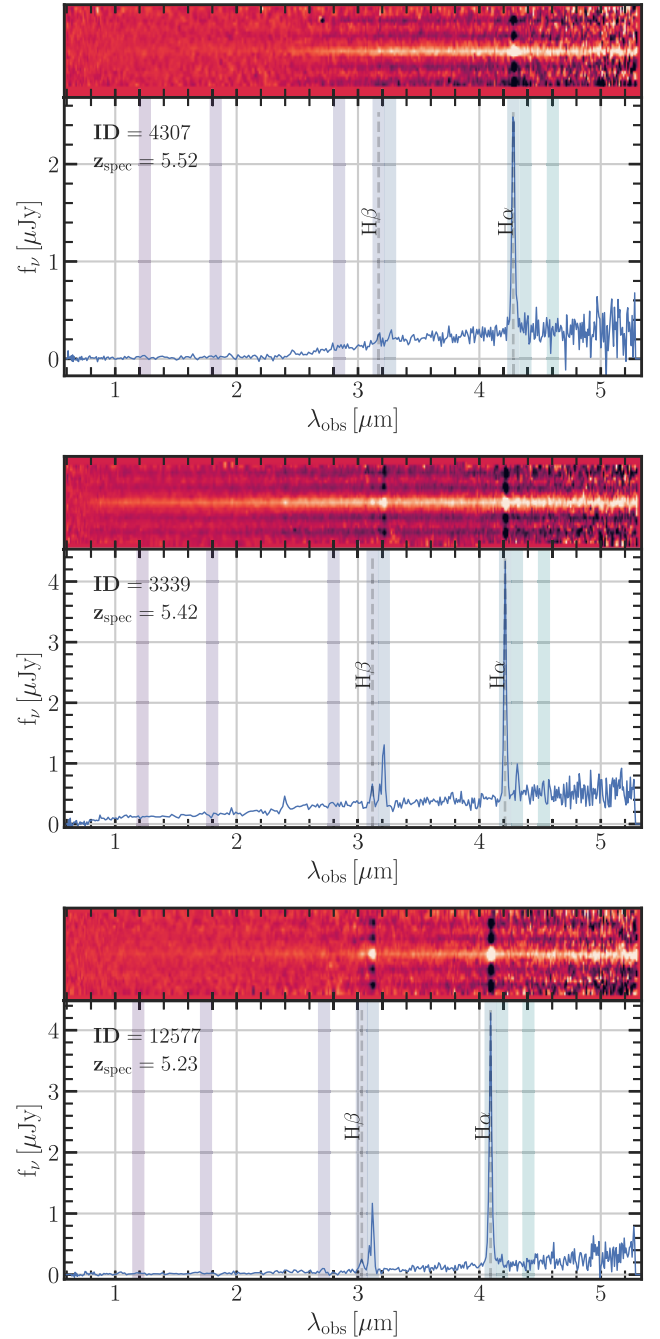


Figure 6. The spectra of the three highest redshift red galaxies in our sample, with redshifts of $z = 5.52, 5.41$ and 5.23 . They exhibit flat spectra featuring pronounced $H\alpha$, $H\beta$, and [O III] emission lines. These spectral characteristics enable precise determination of dust attenuation via the Balmer decrement, circumventing the reliance on upper limits.

uncovers a $z_{\text{spec}} = 3.5$ galaxy devoid of emission lines, showing that it is a quiescent galaxy. The Balmer break is less intense than in the other two quiescent galaxies.

Despite the limited sample size precluding definitive conclusions on galaxy evolution, the spectroscopic identification of three quiescent galaxies at $z > 3$, one exhibiting clear AGN activity, raises questions about diverse quenching mechanisms. PRISM spectra suffice for redshift determination and AGN detection via the $H\alpha$

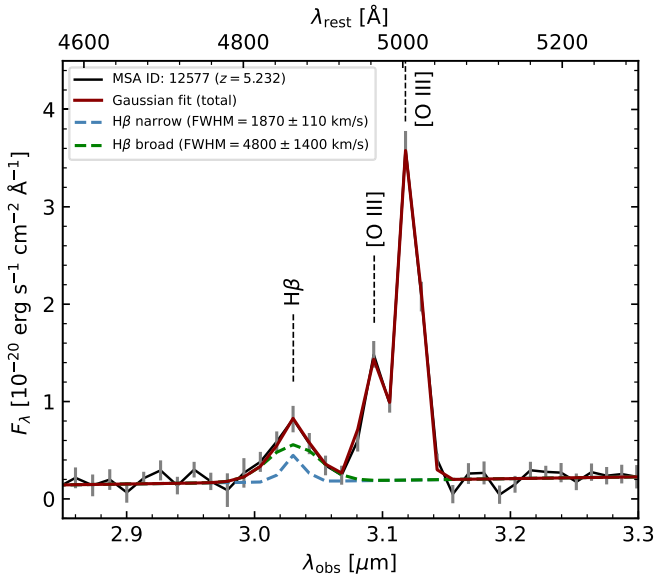


Figure 7. Spectral fitting for the ID = 12577, the only source in the sample that shows another broad line indicating strong evidence of an AGN. The black line is the spectra, whereas the red lines show the fitting for H β , [O III] and the continuum. The narrow component of the H β line is shown in blue, and the broad component is in green.

line in these galaxies. However, these results underscore the need for more extensive and higher-resolution spectroscopic surveys to provide deeper insights into the quenching processes in galaxies of this kind.

4.3 Morphology of red galaxies

This section explores the varied morphologies of red galaxies, linking their morphological features with spectral characteristics, a connection further elaborated in forthcoming work by Baggen et al. (in preparation).

Fig. 9 showcases the RGB images within the main red galaxy subsample, categorized based on spectral analysis into extremely red galaxies, quiescent galaxies, and flat spectra (Figs 5, 6–8). RGB images were generated using the F200W, F356W, and F444W filters as blue, green, and red channels. For IDs 1548 and 12577, lacking F356W coverage, the F444W images were adjusted to match the F356W flux levels expected from their SED fits. The F444W images were analysed with GALFIT to fit Sérsic profiles, characterized by a Sérsic index n and an effective radius R_e [see Baggen et al. (in preparation) for methodology details].

The galaxies with redder spectra typically present more expansive structures, with effective radii exceeding 2 kpc, which is over twice the size of the quiescent galaxies in our data set, whose effective radii range from $0.66 < r_e/\text{kpc} < 0.76$. Morphologically, the three quiescent galaxies exhibit a consistent structure, with an effective radius of $r_e \sim 0.7 \text{ kpc}$ in the F444W filter. Notably, galaxy ID = 8777 contains a central AGN, yet its effective radius remains in line with the other quiescent galaxies in our sample. In contrast, galaxies with $z_{\text{spec}} > 5$, one of them confirmed as an AGN, manifest as compact point sources. We find that the effective radii are constrained to $r_e < 1 \text{ kpc}$. Despite the limited sample size, this result suggests that morphology can offer insights into the underlying nature of red galaxies (further details in Baggen et al. in preparation).

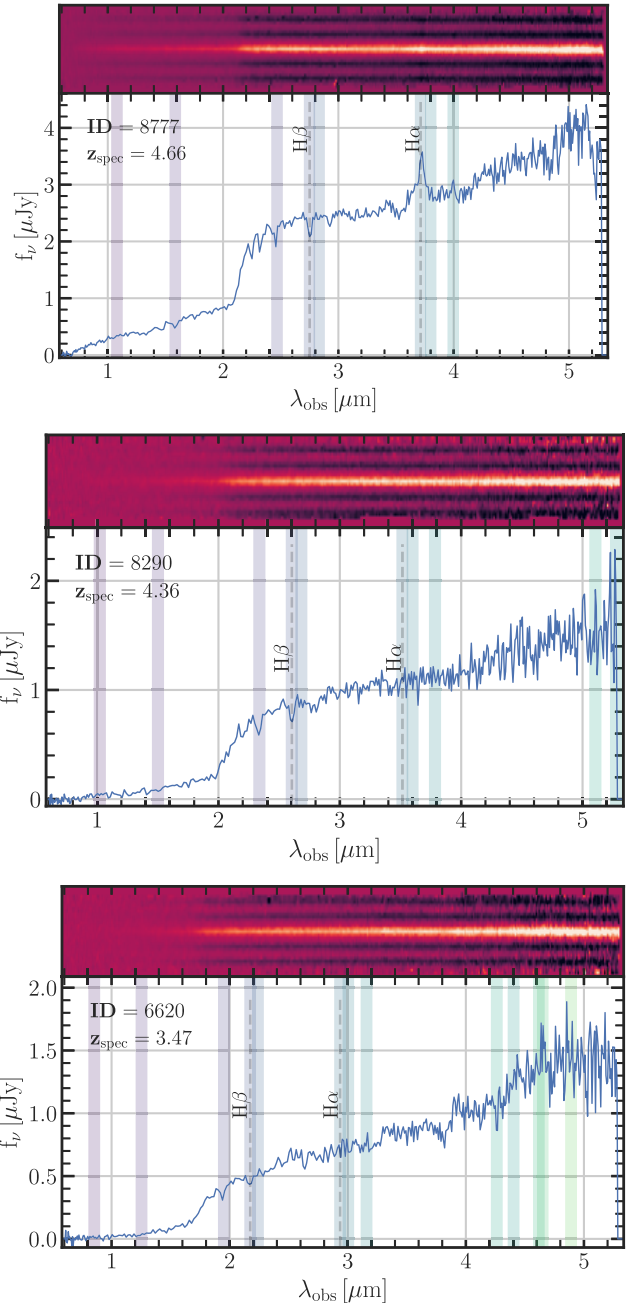


Figure 8. 2D and 1D spectra of the three quiescent galaxies of our red galaxy sample at $z_{\text{spec}} = 4.7$, 4.47 , 3.6 , respectively. The spectra are dominated by strong continuum breaks near $\sim 2 \mu\text{m}$ that are identified as the redshifted Balmer/4000 Å break (dashed lines show H α and H β , and the coloured lines represent [O III], [S II], He I, and [C I], respectively). The first spectra likely present a broad H α line (which is blended with [N II]), whereas the other two lack this property. These quiescent galaxies, making up 13^{+9}_{-6} per cent of red galaxies in our sample, underscore the presence of inactive galaxies within the dusty red population at $z > 3$.

5 PHOTOMETRIC RESULTS: ARE RED GALAXIES EXCEPTIONALLY MASSIVE?

This section presents the physical properties of 19 red galaxies, including SFRs, stellar masses, and dust attenuation. These parameters were derived using the photometric catalogue detailed in Section 2.

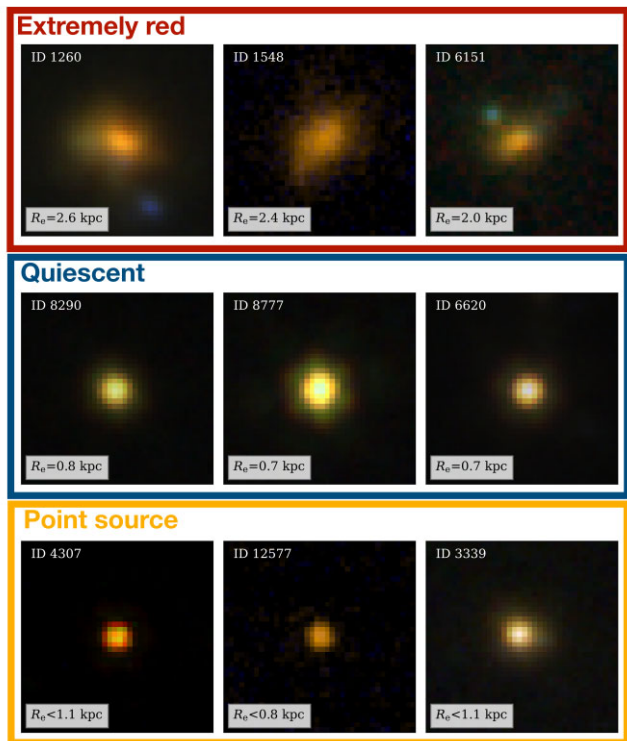


Figure 9. RGB images of main red galaxies subsamples generated using the F200W, F356W, and F444W filters (blue, green, and red colours, respectively). The first row features extremely red galaxies (IDs: 6151, 1260, and 1548), the second showcases quiescent galaxies (IDs: 8290, 8777, and 6620), and the third displays the highest redshift of our sample classified as flat spectra, one of them showing broad lines and classified as AGN (IDs: 4307, 3339, and 12577). Red galaxies present distinct morphologies; the extremely red galaxies are more extended, with an effective radius of approximately $r_e \sim 2.5$ kpc, in contrast to the quiescent galaxies, which are more compact with an effective radius around $r_e \sim 0.7$ kpc. The point sources present effective radius constrained up to $r_e \lesssim 1$ kpc.

We further incorporate spectroscopic redshift data as a pivotal input, which allows us to avoid degeneracies and obtain better physical parameter constraints (see Section 3 for methodology details). The analysis encompasses both quiescent and dust-enshrouded galaxies alongside bluer galaxies that deviate from the established red colour criterion. This comparison enables us to juxtapose red galaxies with a spectroscopic-matched sample from identical observations.

In Fig. 10, we present the spectroscopic redshift distribution of red galaxies versus their stellar mass. Red galaxies are located at $z_{\text{spec}} = 4.1 \pm 0.7$, as detailed in Section 4. These galaxies emerge as the most massive in our samples, with stellar masses ranging from $9.1 < \log(M_*/M_\odot) < 10.8$ with a mean of $\log(M_*/M_\odot) = 10.1 \pm 0.4$. Notably, the redder galaxies, which have masses around $\log(M_*/M_\odot) = 10.5 \pm 0.2$, are comparable to quiescent galaxies at similar redshifts ($z_{\text{spec}} \sim 4.4$). The highest redshift galaxies at $z_{\text{spec}} > 5$ present more moderate stellar masses, $\log(M_*/M_\odot) \sim 10$.

Red galaxies exhibit a broad range in SFRs, with a mean of $\text{SFR} = 59 \pm 44 M_\odot \text{ yr}^{-1}$, yet the dustier subset consistently shows $\text{SFR} < 140 M_\odot \text{ yr}^{-1}$. This is consistent with the broad range of equivalent widths ($70 \text{ \AA} < \text{EW} < 550 \text{ \AA}$) reported in the spectral analysis (refer to Section 4). This indicates that red galaxies may exhibit varied specific star formation. Specifically, sources with high equivalent widths ($\text{EW} \sim 500 \text{ \AA}$), such as IDs 4820 and 4490, suggest starburst

activity. Yet, the average $\text{EW} \sim 300 \text{ \AA}$ across our sample aligns with the main sequence (MS) of galaxies (Mármol-Queraaltó et al. 2016). SED analysis further supports this, indicating moderate SFRs and significant stellar masses, consistent with MS characteristics rather than starburst classification. It is important to note that although this study incorporates total SFRs from BAGPIPES, it does not include far-infrared/submillimetre photometry, which may affect the SFR estimations obtained from SED fitting. None the less, Williams et al. (2024) demonstrated that omitting ALMA data does not significantly alter SFRs for a similar source selection at $z_{\text{phot}} < 5$.

For the dusty galaxies, we explore the well-documented correlation between stellar mass and dust attenuation in dusty galaxies (Genzel et al. 2015; Whitaker et al. 2017; Fudamoto et al. 2020). While the interdependence of these parameters is established, some research suggests this relationship may not evolve beyond $z > 2$ (McLure et al. 2018). In addition, recent findings by Gómez-Guijarro et al. (2023) confirm a $M_{\text{star}}-A_V$ correlation specifically for red galaxies finding a different relation between red galaxies and Lyman Break Galaxies. Our analysis extends this work by quantifying dust attenuation for red galaxies and bluer, less massive counterparts at $z > 3$, and uniquely, we compute dust attenuation using continuum measurements and the Balmer decrement for both cases. Fig. 11 juxtaposes the stellar masses and dust attenuation of both red galaxies and also the bluer galaxies that do not satisfy the red colour criterion ($H-F444W < 1.75$). Our analysis indicates that red galaxies are more massive and dustier than their bluer counterparts. Red galaxies exhibit a different mass–dust attenuation correlation than the bluer galaxies, which follow a similar trend to (McLure et al. 2018), for both SED-based dust attenuation measurements and nebular emission. The dust attenuation values derived from the Balmer decrement exceed those obtained from the continuum (refer to Section 4) and suggest an alternate correlation with stellar mass.

In summary, our results confirm that red galaxies are uniquely massive and dust-rich, differing significantly from their bluer, less massive counterparts. However, we do not find that red galaxies have masses of extremely massive galaxies exceeding $10^{11} M_\odot$.

6 DISCUSSION: IS IT TIME TO REEVALUATE RED COLOUR SELECTIONS IN THE JWST ERA?

This section discusses the nature of red galaxies, contrasting our results with those of earlier studies. We also explain our study’s caveats and propose future improvements.

As a first caveat, we notice that the quiescent subset within red galaxies exhibits considerable uncertainty (13^{+9}_{-6} per cent), and a larger sample would be needed to improve statistics. In addition, notice that there are biases introduced by the NIRS spec mask weighting that should warrant consideration. Utilizing NIRCcam data, Pérez-González et al. (2023) proposed a ‘trifinity’ in red galaxies, identifying ~ 18 per cent as quiescent. While this estimate is consistent with our observations, we consider it an upper limit and advocate for more extensive spectroscopic data. Despite the limited size of our sample precluding definitive conclusions about the UVJ diagrams’ reliability, our study’s highest redshift quiescent galaxy is categorized within the star-forming region according to Carnall et al. (2020). Although UVJ diagrams are extendedly used for classifying quiescent galaxies (Williams et al. 2014; Valentino et al. 2020, 2023; Carnall et al. 2023a), this discrepancy highlights the need to investigate alternative methods to UVJ diagrams for identifying quiescent galaxies, as suggested in the works of Carnall et al. (2018, 2020, 2023a).

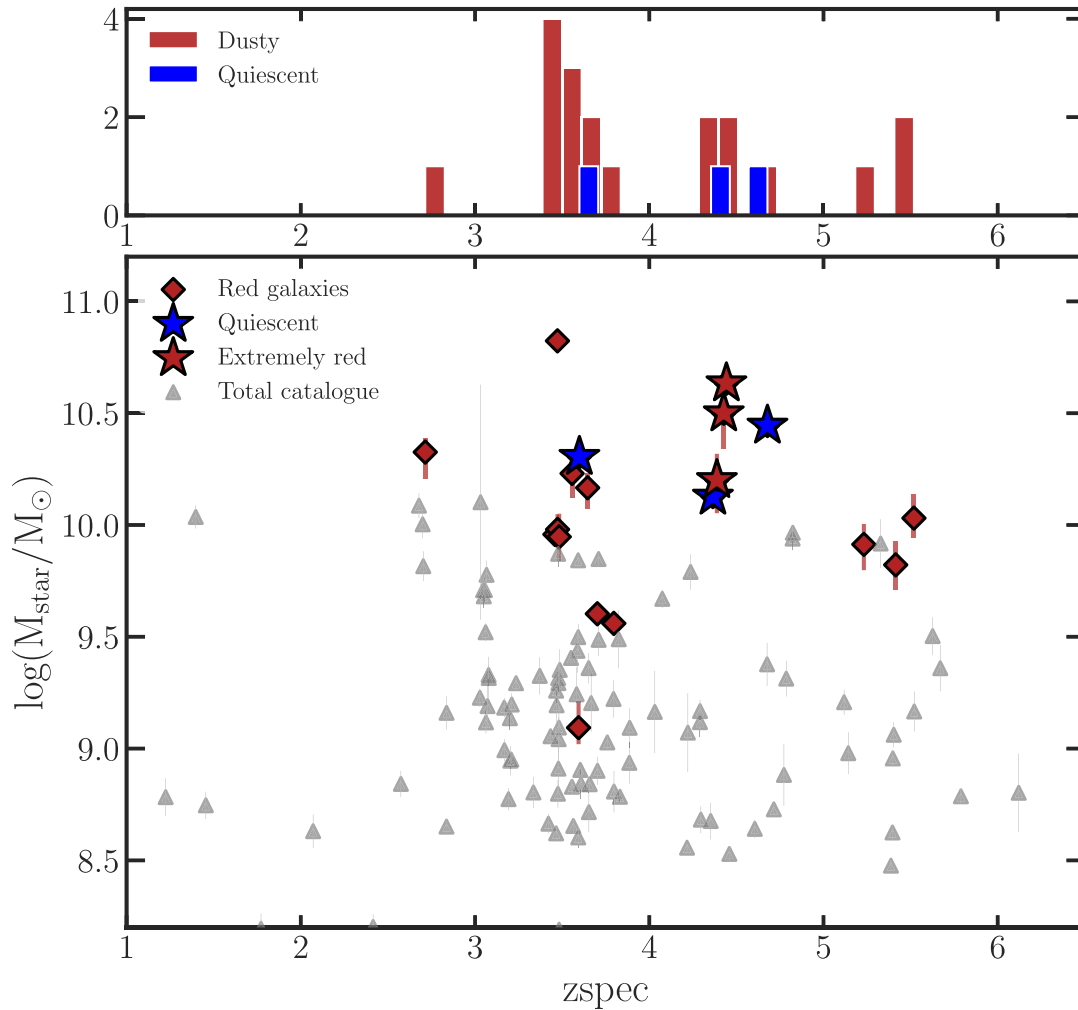


Figure 10. Spectroscopic redshift and stellar mass distribution of red galaxies. The upper diagram displays a histogram of the spectroscopic redshift for dusty (in red) and quiescent (in blue) galaxies. The lower plot displays the stellar mass versus the spectroscopic redshift, with red galaxies represented by red diamonds. The red stars show the most extreme red galaxies, and the blue stars show the quiescent galaxies. The remainder of the sample, the fillers in the ID 2198 Cycle 1 programme, is indicated by grey triangles. Red galaxies primarily lie at $z_{\text{spec}} > 3$ and are among the most massive galaxies in the sample. Interestingly, the three reddest galaxies have similar redshifts and stellar masses compared to the most distant quiescent galaxies.

Following the nature of red galaxies, we find that most are dusty but have a significant disparity between spectra. Some galaxies (IDs = 6151, 1548, and 1260; red stars along the paper) have extremely red spectra. These three galaxies all lie at $z_{\text{spec}} \sim 4.4$, yet we discount the possibility of an overdensity at this redshift based on their spatial distribution and previously identified overdensities in GOODS-S at higher redshifts (Helton et al. 2024). The pronounced redness does not directly correlate with dust attenuation, as these galaxies exhibit varied levels of dust attenuation. In a related finding, Arrabal Haro et al. (2023) presented a galaxy with a similar red spectrum at $z_{\text{spec}} = 4.9$, initially reported in Barrufet et al. (2023a). This pattern may indicate a selection bias wherein redder colours are more likely to identify the most massive galaxies, akin to the ‘*Spitzer HST-dark*’ galaxies described by Wang et al. (2019).

Pérez-González et al. (2024) identified highly attenuated sources ($A_v \sim 10$ mag) at $z_{\text{spec}} > 5$, suggesting a combination of star formation and AGN activity in compact red galaxies (with the characteristic V-shape), the so-called little red dots. Conversely, Greene et al. (2024), Kocevski et al. (2023), and Matthee et al. (2024) support that

these compact sources are AGN. Furthermore, the LRD definition has evolved from its original selection criteria, with various authors adopting disparate definitions for these sources. Our data indicate at least one compact point source at $z_{\text{spec}} > 5$ is an AGN, with lower attenuation levels than Pérez-González et al. (2024) ($A_v \lesssim 5$ mag). We also highlight the issue of blending $H\alpha + [N\text{II}]$ in low-resolution NIRSpect spectra, which could inflate dust attenuation estimates, potentially reducing actual attenuation to $A_v \sim 2$ mag if $[N\text{II}]$ flux contributes significantly (up to 40 per cent).

The discovery of large stellar masses in red galaxies by initial *JWST* NIRCcam studies (Labbé et al. 2023) and subsequent confirmation with FRESCO data (Xiao et al. 2024) have sparked considerable discussion. Contrarily, some works suggest that incorporating MIRI data might lead to lower stellar mass estimates (Williams et al. 2024; Pérez-González et al. 2024), a point recently emphasized by Wang et al. (2024). Moreover, Barro et al. (2024) revealed that stellar mass estimates could vary by approximately 1 dex based on the choice of SED-fitting code. In a distinct approach, Lu et al. (2024) reassessed stellar population models, initially tested on a limited set

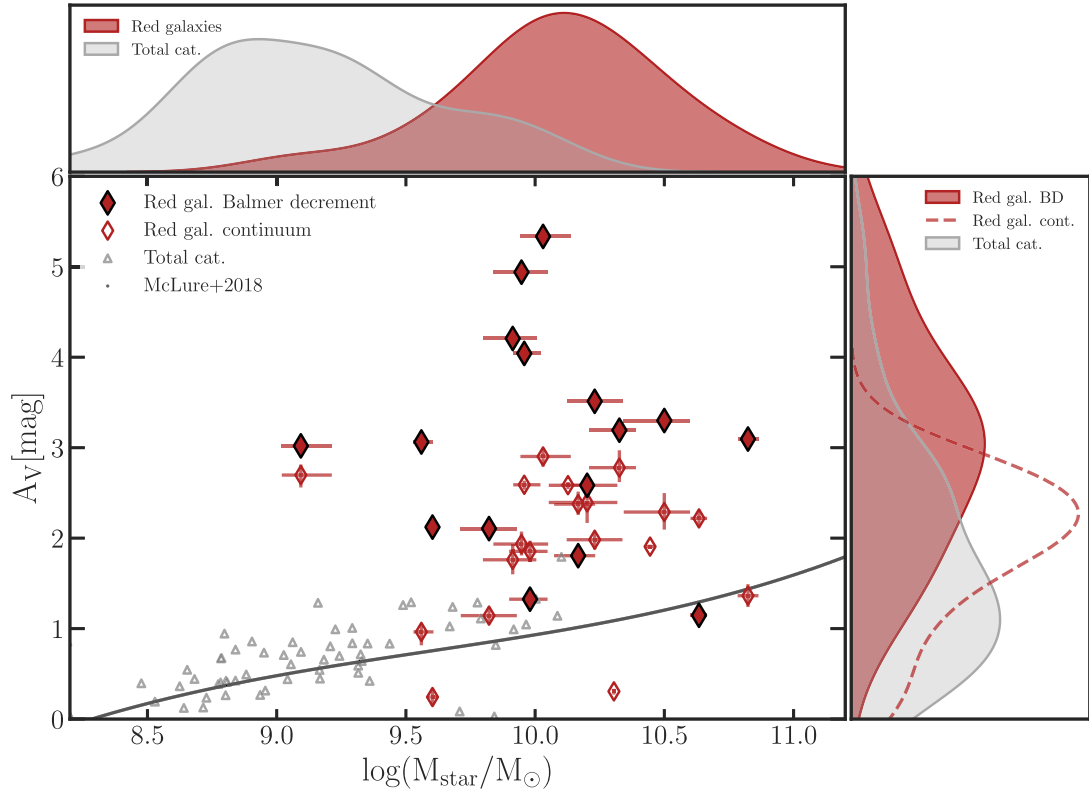


Figure 11. Comparison of stellar mass and dust attenuation (A_V) for red galaxies and bluer, non-red counterparts. Open symbols denote dust attenuation inferred from the continuum, whereas solid symbols indicate values obtained through Balmer decrement analysis. The black dots represent the correlation from McLure et al. (2018), aligning with the bluer galaxies in our sample. Kernel density estimation (KDE) diagrams adjacent to the main plot (top and right) contrast the distribution of red galaxies (in red) with that of the broader sample (in grey). Attenuation values obtained through the Balmer decrement consistently exceed those derived from the continuum, exhibiting a steeper correlation.

of galaxies, raising the question of their potential impact on the mass estimates of red galaxies. Although various factors could influence stellar mass determinations, extending beyond this paper’s focus, our analysis indicates that our sample of red galaxies at $z_{\text{spec}} > 5$ do not exhibit anomalously high masses, aligning with current cosmological models.

This work demonstrates, via the Balmer decrement, that ~ 90 per cent of red galaxies are very dusty. However, we cannot assess the total obscured star formation since the absence of far-infrared and submillimetre data limit us. To accurately determine the total SFR, it is essential to incorporate ALMA data. Wang et al. (2019) have provided observations for three galaxies within this sample. However, the rest of this sample has been unexplored so far. Concurrently, several studies show the serendipitous dust continuum detections of UV-bright galaxies (Fudamoto et al. 2021; Barrufet et al. 2023b), altogether suggesting the need for deeper observations. Future ALMA cycles will be critical in fully characterizing red galaxies.

Our study reveals that red galaxies – a term encompassing H-dropouts, *HST*-dark, and closely related to *HST*-faint, optically faint, optically invisible galaxies – span diverse characteristics, including quiescent, dusty, and potentially AGN-influenced nature, alongside variations in morphology and mass. This diversity prompts the consideration of whether, in the *JWST* era, we should shift towards selecting galaxies based on physical properties (i.e. mass and redshift). While such selections pose challenges, the accumulated knowledge and the integration of *JWST* instruments like NIRCarn, NIRSpec, and

MIRI in extensive multiband surveys suggest a possible transition towards physical properties selection criteria.

7 SUMMARY AND CONCLUSIONS

Through the 10 h Cycle-1 programme (ID 2198, PIs Barrufet, Oesch) titled ‘Quiescent or dusty? Unveiling the nature of extremely red galaxies at $z > 3$ ’, we have analysed the spectra of 23 red galaxies ($H\text{-F444W} > 1.75$). While these galaxies are generally presumed to be dusty, there has been speculation about the presence of quiescent galaxies among them, as photometric data alone, even from NIRCarn data, cannot reliably differentiate between the two galaxy populations. Our programme shows that low-resolution NIRSpec/PRISM spectra can discern between quiescent and dusty galaxies.

The main results of the paper are as follows:

- (i) The mean spectroscopic redshift of red galaxies is $z_{\text{spec}} = 4.1 \pm 0.7$ with the entire sample lying at $z_{\text{spec}} \gtrsim 3$.
- (ii) Red galaxies are mostly dusty, characterized by a dust attenuation from the Balmer decrement of $A_V = 3.1 \pm 1.2$, exceeding the dust attenuation values obtained from continuum measurements, which stand at $A_V = 1.9 \pm 0.8$.
- (iii) Dusty red galaxies are $H\alpha$ emitters with equivalent widths ranging from $68 \text{ \AA} < \text{EW}_{H\alpha} < 550 \text{ \AA}$ suggesting a wide range of recent star formation.
- (iv) Among red galaxies, hidden gems are discovered: quiescent galaxies at $z_{\text{spec}} > 3.5$. Our programme enables us to estimate

that 13 per cent of red galaxies are quiescent. Particularly, the highest redshift ($z_{\text{spec}} = 4.7$) passive galaxy has a broad line of FWHM = 5523 km s⁻¹ unequivocally containing an AGN.

(v) Morphological analysis of red galaxies reveals size variations correlating with spectral types: extremely red galaxies are more extended, averaging $R_e \sim 2.4$ kpc, quiescent galaxies are compact, typically around $R_e \sim 0.7$ kpc. The three very compact sources at $z_{\text{spec}} > 5$ present radius $R_e < 1$ kpc, but we only confirm AGN in one of them.

(vi) Red galaxies are predominantly massive with ~ 85 per cent of the sample having stellar masses of $\log(M_*/M_\odot) \geq 9.8$. Notably, the extremely red galaxies are among the most massive red galaxies ($\log(M_*/M_\odot) = 10.5 \pm 0.2$). These masses are on par with those of the quiescent red galaxies in this sample at similar redshifts $z_{\text{spec}} \sim 4.4$.

Our study has significantly advanced our understanding by showing that red galaxies constitute a non-homogeneous population. While colour is a reliable indicator of mass, it falls short of fully revealing the nature of red galaxies. The trend observed is that the redder galaxies are the most massive, encompassing quiescent and dusty types.

This pilot programme has effectively shown the NIRSpec/PRISM's ability to distinguish between quiescent and dusty galaxies and to shed light on dust attenuation via spectral analysis. None the less, more extensive surveys with higher-spectral resolution are required for precise attenuation quantification and exploration of the metallicity–mass relationship in these galaxies. Additionally, the potential synergies between *JWST* and ALMA are yet to be fully leveraged. While this investigation underscores NIRSpec/PRISM's capabilities, a thorough understanding of dust properties in red galaxies necessitates combined analyses with ALMA's submillimetre observations.

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Facilities: *JWST*, *HST*

Software: MATPLOTLIB (Hunter 2007), NUMPY (Oliphant 2015), SCIPY (Virtanen et al. 2020), JUPYTER (Kluyver et al. 2016), ASTROPY (Astropy Collaboration 2013, 2018), GRIZLI (Brammer 2018), SEXTRACTOR (Bertin & Arnouts 1996), BAGPIPES (Carnall et al. 2018)

DATA AVAILABILITY

The data products are available from the authors upon reasonable request.

REFERENCES

- Akins H. B. et al. 2023, *ApJ*, 956, 61
 Alcalde Pampliega B. et al., 2019, *ApJ*, 876, 135
 Arrabal Haro P. et al., 2023, *Nature*, 622, 707
 Astropy Collaboration, 2013, *A&A*, 558, A33
 Astropy Collaboration, 2018, *AJ*, 156, 123
 Barro G. et al. 2024, *ApJ*, 963, 128
 Barrufet L. et al., 2023a, *MNRAS*, 522, 449
 Barrufet L. et al., 2023b, *MNRAS*, 522, 3926
 Bertin E., Arnouts S., 1996, *A&AS*, 117, 393
 Boylan-Kolchin M., 2023, *Nat. Astron.*, 7, 731
 Brammer G., 2018, *Gbrammer/Grizli: Preliminary Release*. Zenodo, <https://zenodo.org/records/1146905>
 Bruzual G., Charlot S., 2003, *MNRAS*, 344, 1000
 Calzetti D., Armus L., Bohlin R. C., Kinney A. L., Koornneef J., Storchi-Bergmann T., 2000, *ApJ*, 533, 682
 Caputi K. I. et al., 2012, *ApJ*, 750, L20
 Carnall A. C., McLure R. J., Dunlop J. S., Davé R., 2018, *MNRAS*, 480, 4379
 Carnall A. C. et al., 2020, *MNRAS*, 496, 695
 Carnall A. C. et al. 2023a, *Nature*, 619, 716
 Carnall A. C. et al., 2023b, *MNRAS*, 520, 3974
 Domínguez A. et al., 2013, *ApJ*, 763, 145
 Eisenstein D. J. et al., 2023a, preprint (arXiv:2306.02465)
 Eisenstein D. J. et al., 2023b, preprint (arXiv:2310.12340)
 Ferland G. J. et al., 2017, *RMxAA*, 53, 385
 Feroz F., Hobson M. P., Cameron E., Pettitt A. N., 2019, *Open J. Astrophys.*, 2, 10
 Franco M. et al., 2018, *A&A*, 620, A152
 Fudamoto Y. et al., 2020, *A&A*, 643, A4
 Fudamoto Y. et al., 2021, *Nature*, 597, 489
 Genzel R. et al., 2015, *ApJ*, 800, 20
 Gómez-Guijarro C. et al., 2023, *A&A*, 677, A34
 Gottumukkala R. et al. 2024, *MNRAS*, 530, 966
 Greene J. E. et al. 2024, *ApJ*, 964, 39
 Gruppioni C. et al., 2020, *A&A*, 643, A8
 Helton J. M. et al. 2024, *ApJ*, 974, 41
 Hunter J. D., 2007, *Comput. Sci. Eng.*, 9, 90
 Kashino D. et al., 2013, *ApJ*, 777, L8
 Kluyver T. et al. IOS Press, 2016, in Loizides F., Schmidt B., eds, Positioning and Power in Academic Publishing: Players, Agents and Agendas. p. 87
 Kocevski D. D. et al. 2023, *ApJL*, 954, L4
 Kroupa P., 2001, *MNRAS*, 322, 231
 Labbé I. et al., 2015, *ApJS*, 221, 23
 Labbé I. et al., 2023, *Nature*, 616, 266
 Lovell C. C., Harrison I., Harikane Y., Tacchella S., Wilkins S. M., 2023, *MNRAS*, 518, 2511
 Lu S. et al. 2024, *Nat. Astron.*,

- Mármol-Queralto E., McLure R. J., Cullen F., Dunlop J. S., Fontana A., McLeod D. J., 2016, *MNRAS*, 460, 3587
- Matthee J. et al. 2024, *ApJ*, 963, 28
- McLure R. J. et al., 2018, *MNRAS*, 476, 3991
- Merlin E. et al., 2019, *MNRAS*, 490, 3309
- Nelson E. J. et al. 2023, *ApJL*, 948, L18
- Oesch P. A. et al., 2023, *MNRAS*, 525, 2864
- Oke J. B., Gunn J. E., 1983, *ApJ*, 266, 713
- Oliphant T. E., 2015, *Guide to NumPy*, Continuum Press
- Pérez-González P. G. et al., 2023, *ApJ*, 946, L16
- Pérez-González P. G. et al. 2024, *ApJ*, 968, 4
- Rodighiero G., Bisigello L., Iani E., Marasco A., Grazian A., Sinigaglia F., Cassata P., Gruppioni C., 2023, *MNRAS*, 518, L19
- Sanders R. L. et al., 2015, *ApJ*, 799, 138
- Santini P. et al. 2021, *A&A*, 652, A30
- Setton D. J. et al. 2024, *ApJ*, 974, 145
- Skelton R. E. et al., 2014, *ApJS*, 214, 24
- Stefanon M. et al. 2021, *ApJSS*, 257, 68
- Valentino F. et al., 2020, *ApJ*, 889, 93
- Valentino F. et al., 2023, *ApJ*, 947, 20
- Virtanen P. et al., 2020, *Nature Methods*, 17, 261
- Wang T. et al., 2016, *ApJ*, 816, 84
- Wang T. et al., 2019, *Nature*, 572, 211
- Wang T. et al., 2024, preprint ([arXiv:2403.02399](https://arxiv.org/abs/2403.02399))
- Weibel A. et al. 2024, *MNRAS*, 533, 1808
- Whitaker K. E., Pope A., Cybulski R., Casey C. M., Popping G., Yun M. S., 2017, *ApJ*, 850, 208
- Williams C. C. et al., 2014, *ApJ*, 780, 1
- Williams C. C. et al., 2019, *ApJ*, 884, 154
- Williams C. C. et al. 2024, , *ApJ*, 968, 34
- Xiao M. et al. 2024, *Nature*, 635, 311
- Xiao M. Y. et al., 2023b, *A&A*, 672, A18
- Yamaguchi Y. et al., 2019, *ApJ*, 878, 73

APPENDIX A: 2D SPECTRA OF OUR SAMPLE

Here, we present the spectra of the 19 red galaxies in our sample, which constitute the main targets of the cycle 1 *JWST*/NIRSpec programme: ‘Quiescent or dusty? Unveiling the nature of red galaxies at $z > 3$ ’ (GO-2198; PIs: L. Barrufet & P. Oesch).

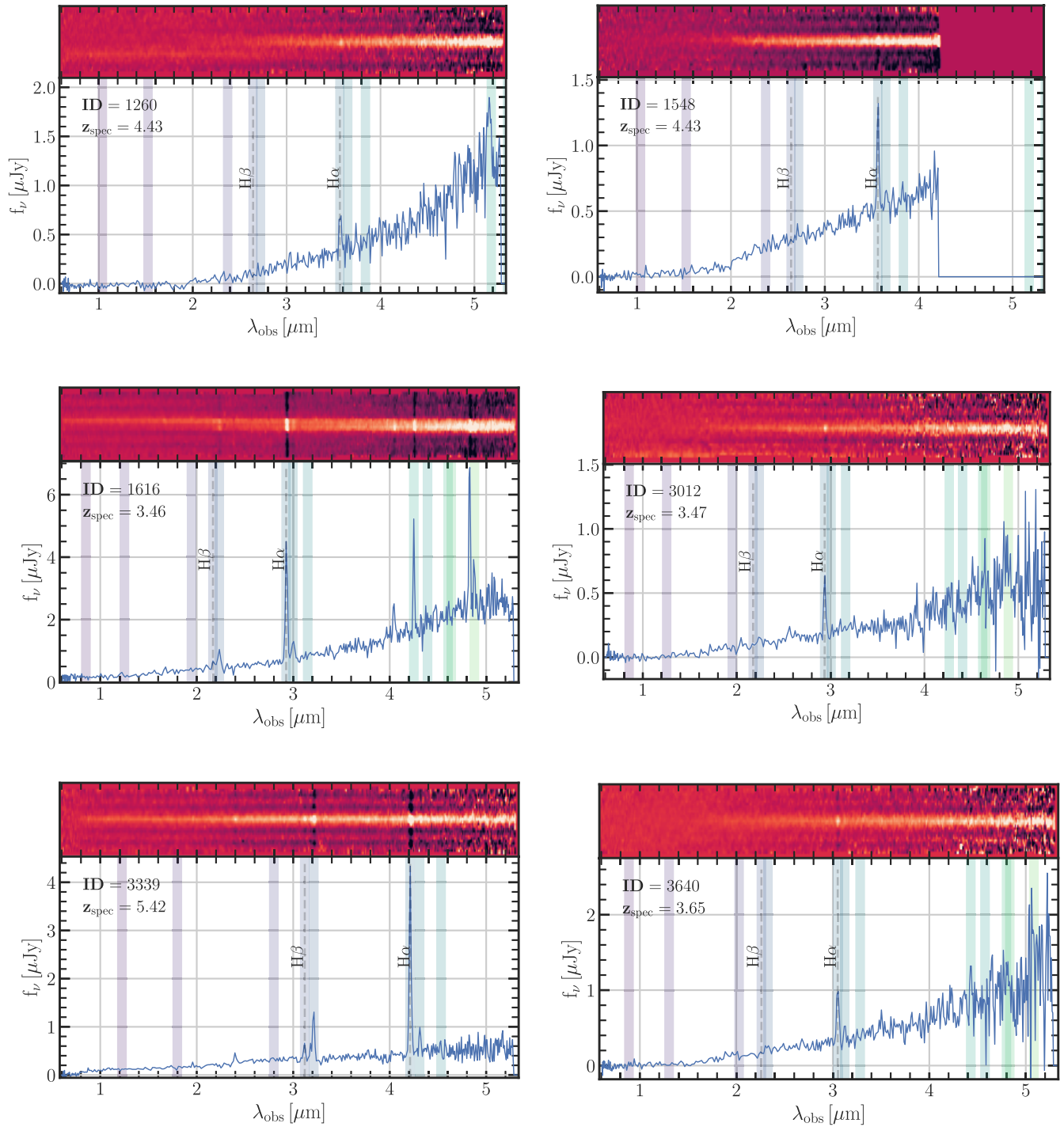
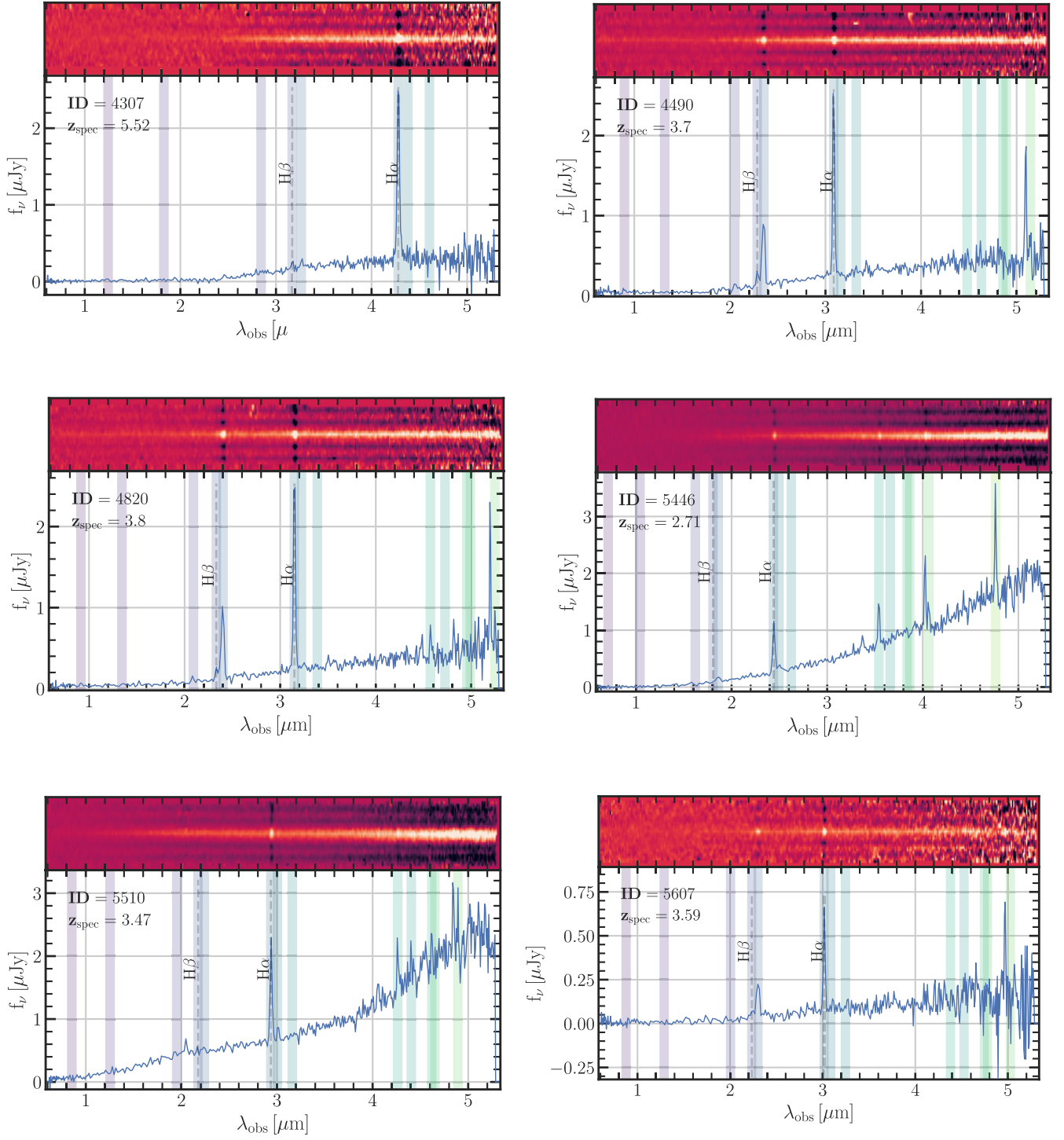


Figure A1. 2D and 1D Spectra for the red galaxies. The dashed lines mark the locations of the H α and H β spectral lines. Coloured lines represent [O III], [S II], He I, and [C I], respectively. The ID and the spectroscopic redshift are shown in the upper left of the 1D plot.

Figure A1. *continued.*

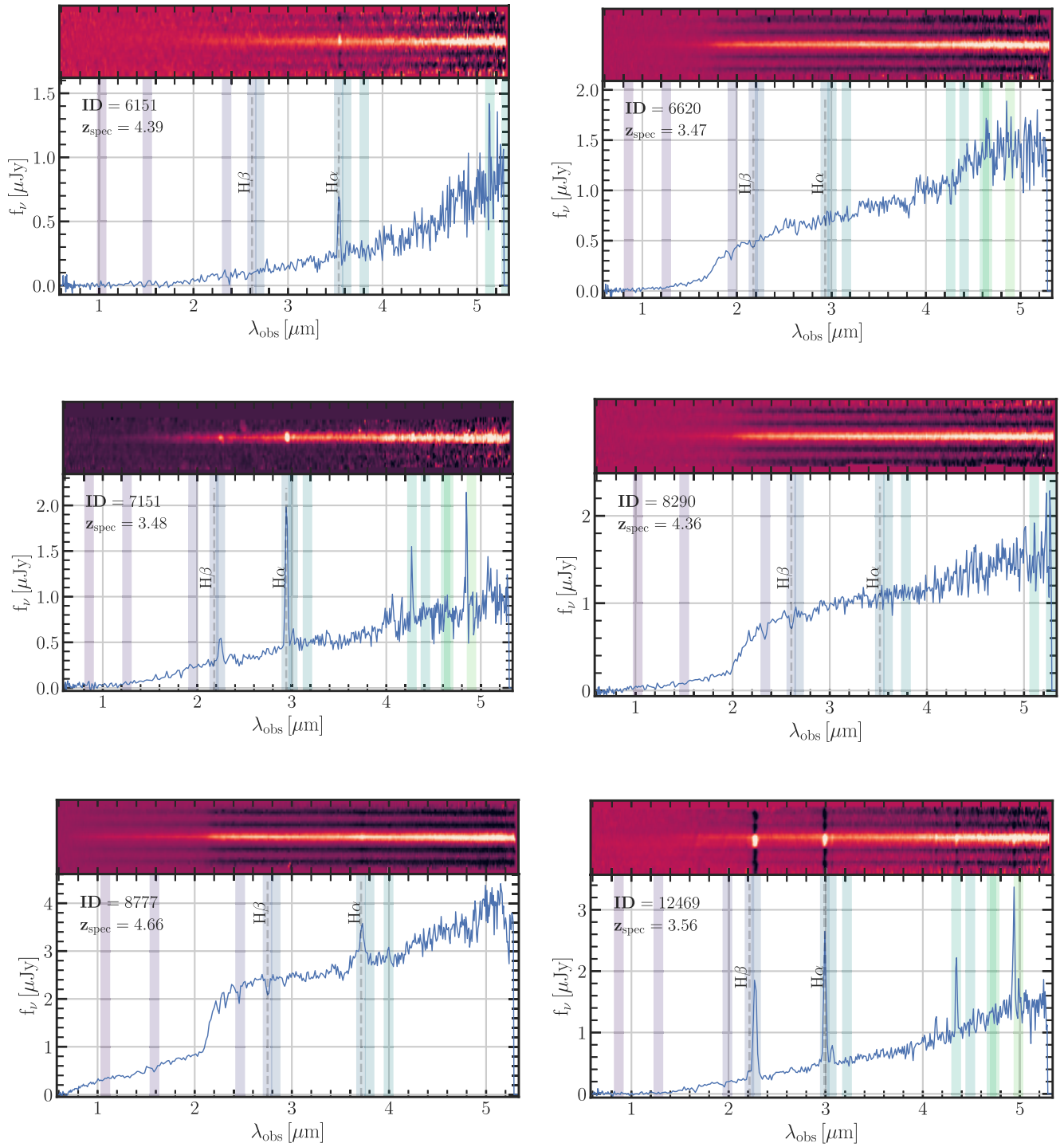


Figure A1. *continued.*

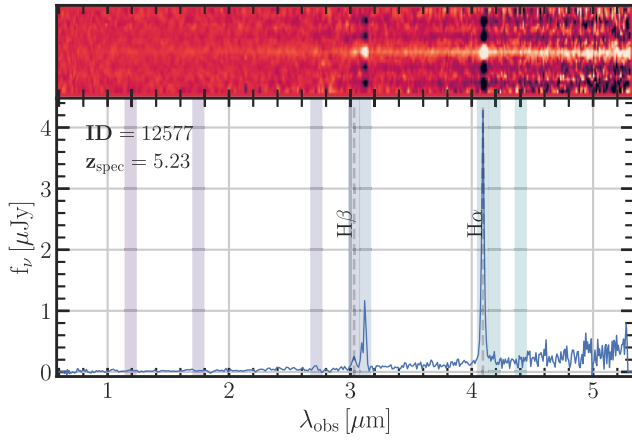


Figure A1. *continued.*

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